

Closing the Pre-Infusion Gap: A Quality Improvement Initiative to Enhance Laboratory Readiness for Outpatient IV Iron Referrals

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Laboratory completion increased from 69.1% to 89.8%

MOTIVATION

At Renown in Northern Nevada, some patients arrived for IV iron infusions without valid iron studies, requiring same-day testing and coordination among nursing staff, pharmacy, and referring clinicians. Lab orders embedded in treatment plans did not reliably ensure readiness before arrival, highlighting the need for earlier verification.

★ **Nearly 1 in 3 lacked valid iron studies before infusion**
Baseline: 64/207 | Jan–Apr 2026

AIM
Increase required iron study completion to ≥85% within six months among internal IV iron referrals, supported by verification before scheduling.

ESTIMATED CHAIR-TIME BURDEN

Baseline period: Jan–Apr 2026
Assuming 75 additional chair-minutes per visit lacking valid iron studies

- 20 additional chair-hours/month
- 240 chair-hours/year (annualized)*

**Based on an infusion pharmacy staff estimate of 75 extra minutes per affected visit. Annual projection assumes unchanged baseline volume and frequency of missing or invalid studies; delays were not directly measured.*

METHODS

Single-center QI evaluation of 416 internal outpatient IV iron referral records, January 1–September 14, 2026. All records were included regardless of treatment status.

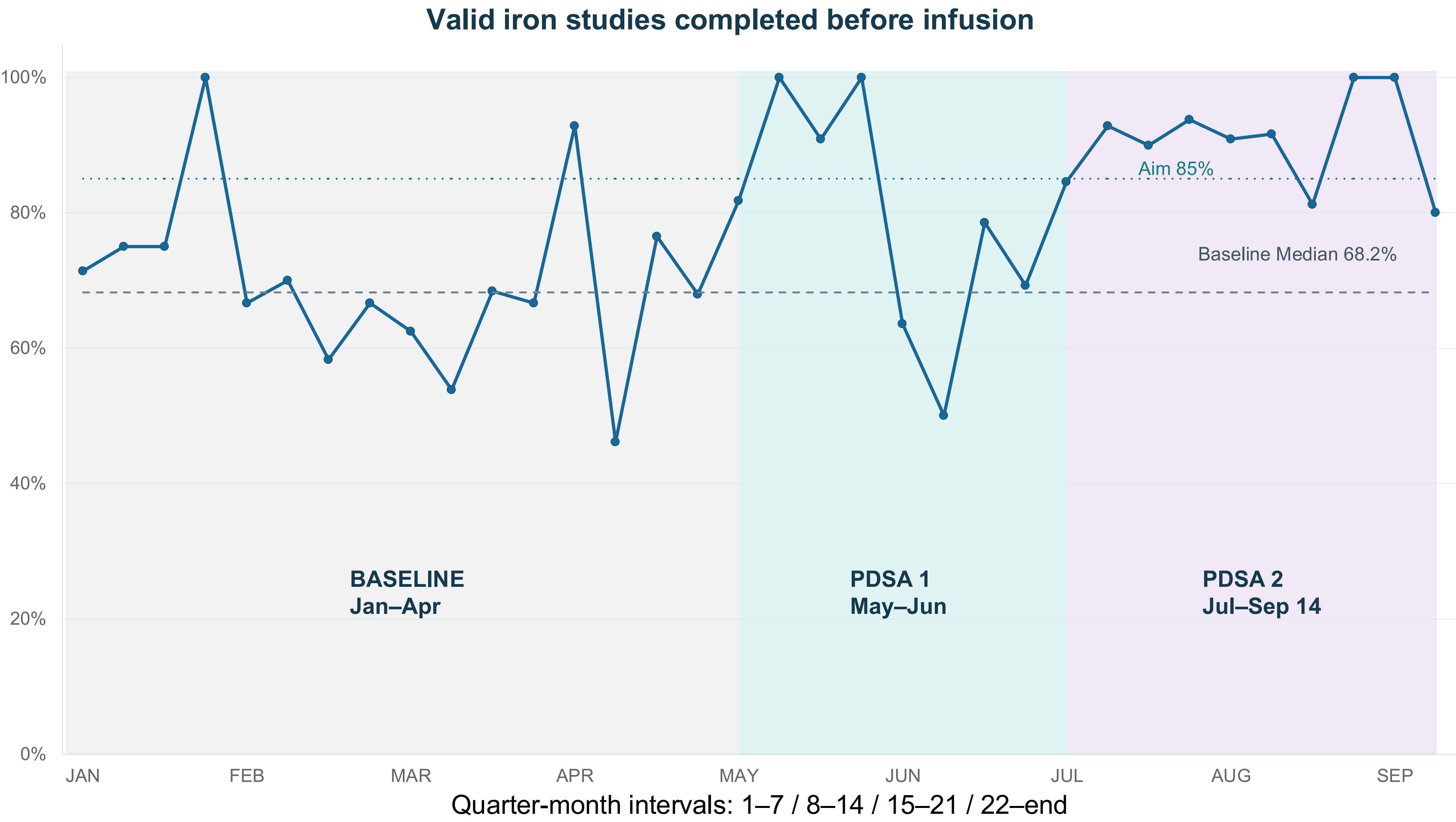
Process measure: Percentage with complete, valid iron studies before infusion.

Outcome measure: Percentage meeting laboratory treatment parameters.

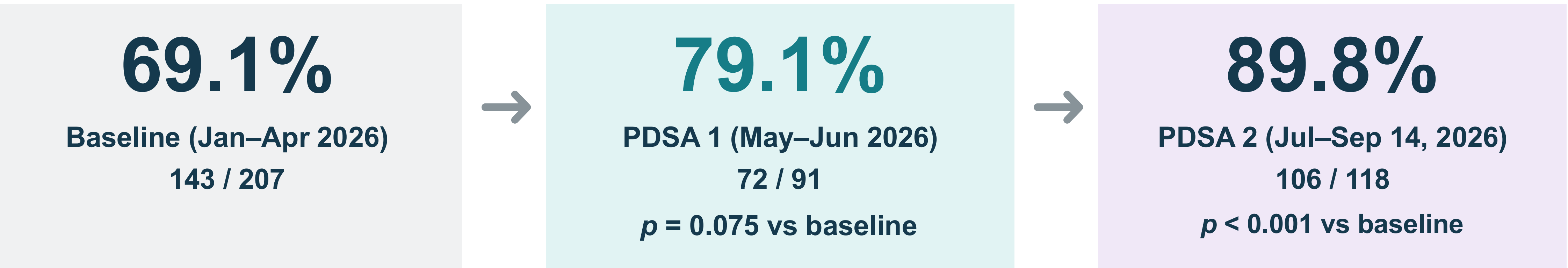
Analysis:

- Quarter-month run chart displayed process performance over time using a fixed baseline median.
- Pearson’s chi-square test compared pooled laboratory readiness rates across the three study periods: baseline, PDSA 1, and PDSA 2, assuming independent records. Outcome rates were summarized descriptively.

PROCESS MEASURE: LABORATORY READINESS OVER TIME

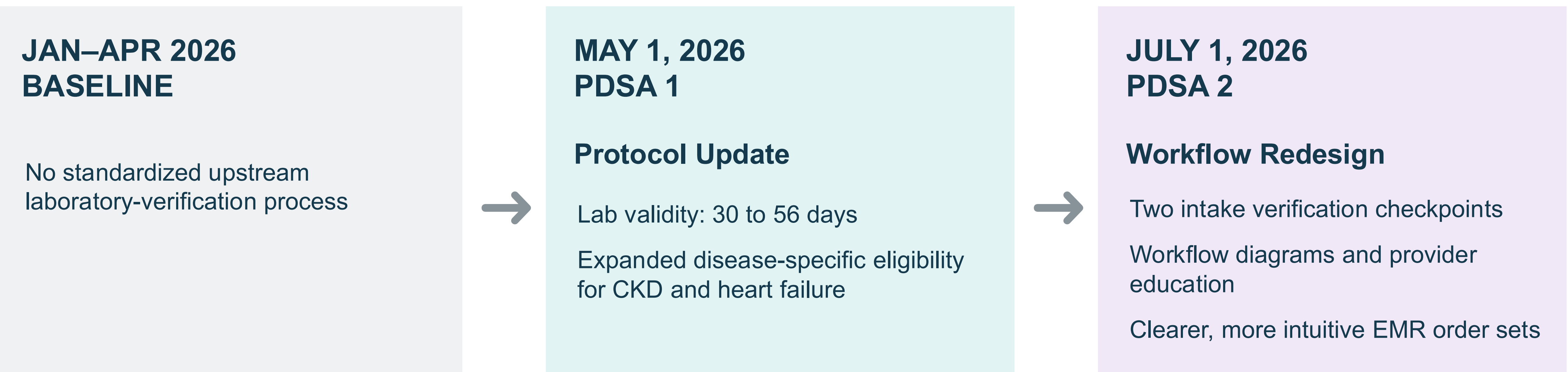


LABORATORY READINESS BY PERIOD



Overall $p < 0.001$ across three periods.
Pairwise p-values compare each intervention period with baseline using Pearson’s chi-square tests with Holm adjustment.

WHAT WE CHANGED



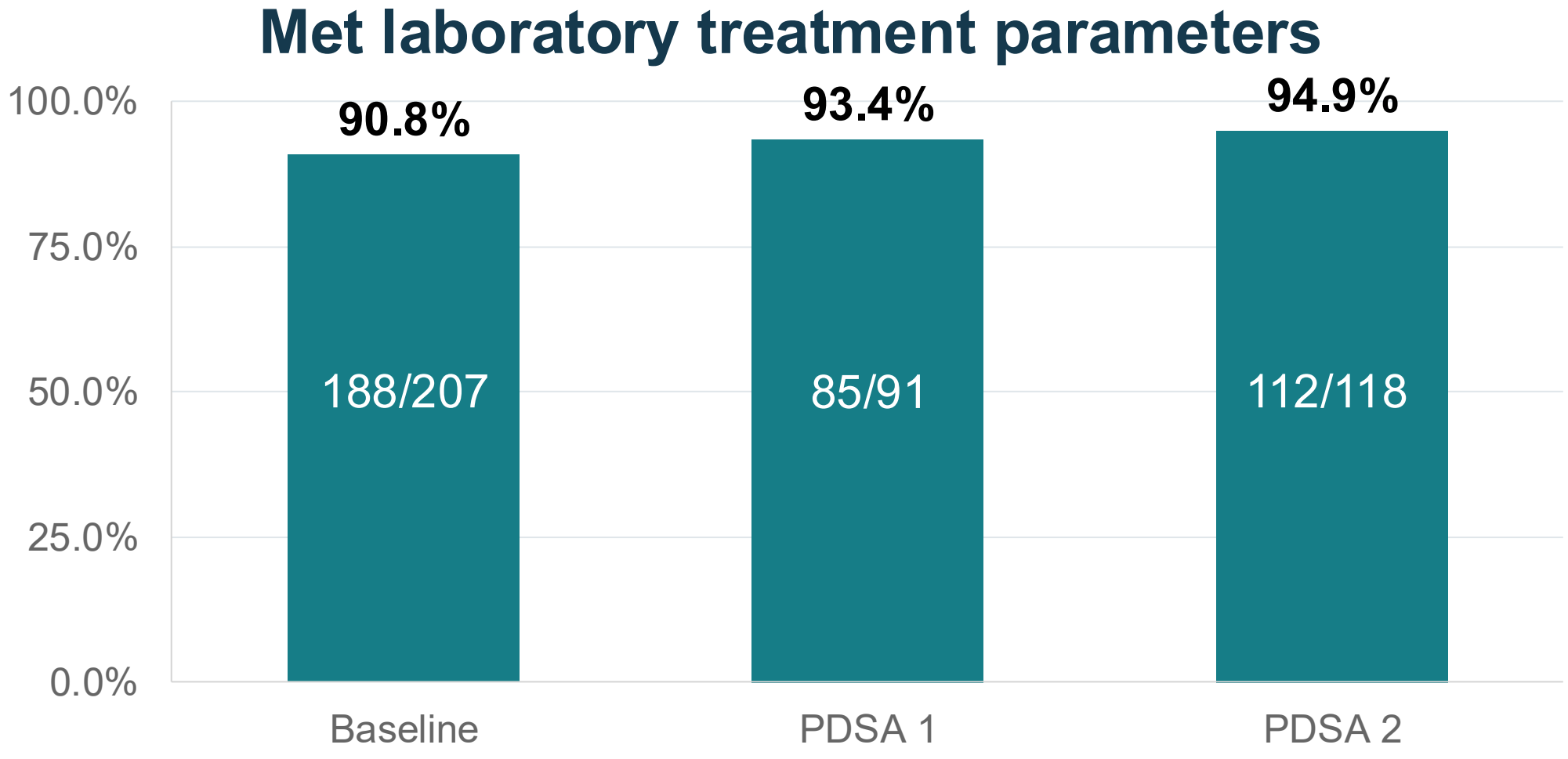
ACKNOWLEDGMENTS

We thank Renown’s infusion nursing, pharmacy, and intake teams for implementation support, and American Society of Hematology mentors Drs. Ming Lim and Richard Godby for their guidance.

REFERENCES

Renown Health. *Pricing Transparency*. Accessed September 30, 2026. <https://www.renown.org/Patients-and-Visitors/Billing/Pricing-Transparency>

OUTCOME MEASURE



Attainment remained high across all periods.

PATIENT IMPACT (ESTIMATED)

≈161 patient-hours of extra chair time potentially avoided annually* from baseline to PDSA 2.

Potentially less waiting and fewer treatment disruptions for patients and caregivers, including those traveling from rural Northern Nevada.

MODELED ANNUAL CAPACITY BURDEN

Annual burden*	Baseline	PDSA 1	PDSA 2
Extra chair-hours	240	162	79
Hypothetical opportunity cost†			
Lowest listed Medicaid	\$14,700	\$9,900	\$4,800
Commercial mean	\$168,100	\$113,500	\$55,300
Hometown Health National	\$279,000	\$188,000	\$92,000

Released chair time could expand access to oncology and other infusion therapies.

** Assumes 621 visits/year, 75 extra minutes per affected visit (staff estimate), and sustained rates; delays and patient benefits unmeasured.*

† Assumes full utilization at one CPT 96365 visit/chair-hour, valued at \$61.38, \$700.41 (unweighted commercial mean), or \$1,161.30 per visit. Gross revenue proxy before delivery costs; not measured costs, lost revenue, or savings.

LIMITATIONS

Single-center, uncontrolled evaluation with limited follow-up. Changes in laboratory validity and eligibility criteria may affect comparisons across periods. Individual intervention effects cannot be isolated. Chair-time and financial estimates are modeled, not directly measured.

CONCLUSION

Protocol and workflow changes were associated with improved laboratory readiness, exceeding the 85% target overall after PDSA 2. Laboratory treatment-parameter attainment remained high. Intermittent below-target intervals highlight the need for continued monitoring and refinement.

NEXT STEPS

Monitor sustainability and adapt the workflow for an external-referral pilot.

